

DETAILED GUIDELINES

Converting AR/VR Research Projects into Scientific Manuscripts

Core expectation

The manuscript must read as a scientific research paper - not as a project report, software development diary, experimental log, or collection of information.

The central question is not "What did we build?" but "What new scientific knowledge, method, evidence, or design insight does this study contribute?"

Purpose: To guide students in restructuring an AR/VR project into a rigorous, evidence-driven manuscript suitable for submission to a high-quality journal.

Important: Following this guide does not guarantee journal acceptance. Journal fit, novelty, experimental quality, writing quality, ethics, and peer-review outcomes will determine publication.

How to Use This Guide

Students should use this document while redesigning the research problem, planning experiments, preparing figures, analysing results, and writing the manuscript. Do not wait until the end of the project to "convert" a report into a paper. The research design and the paper structure must be aligned from the beginning.

Minimum scientific story

Important problem -> evidence from recent literature -> precise research gap -> novel idea -> research questions/hypotheses -> proposed method -> controlled evaluation -> baseline comparison -> statistical evidence -> interpretation -> limitations -> scientific contribution.

1. Understand the Difference: Project Report vs. Scientific Paper
2. Define the Scientific Core Before Writing
3. Title and Keywords
4. Abstract
5. Introduction
6. Related Work and Research Gap
7. Research Questions and Hypotheses
8. Proposed Method / Framework / System
9. Research Methodology and Experimental Design
10. Baselines and Comparative Evaluation
11. Metrics and Measurement Instruments
12. Statistical Analysis
13. Results
14. Ablation, Sensitivity, Robustness and Generalization
15. Discussion
16. Limitations, Threats to Validity and Ethics
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20. References and Literature Quality
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1. Understand the Difference: Project Report vs. Scientific Paper

A project report documents what was designed and implemented. A scientific paper argues for a new contribution and supports that contribution using evidence. Students must remove the chronological "we did this, then this" style unless a sequence is required to reproduce an experiment.

Avoid / Weak	Use / Strong scientific style
First, we installed Unity. Then we imported a 3D model. Next, we created buttons and connected the VR headset. Finally, we tested the application.	This study proposes a performance-aware immersive training framework that adapts task guidance according to user errors and completion behaviour. The framework is evaluated against a fixed-guidance VR baseline using objective performance measures and user-experience metrics.

Your manuscript should NOT be built around software features. It should be built around a scientific claim. Every major section should support that claim.

Do not submit a paper whose main contribution is only:

- "We developed an AR/VR application."
- "We created a virtual environment."
- "We used Unity/Unreal Engine."
- "Users liked the system."
- "The system achieved good results."

2. Define the Scientific Core Before Writing

Before writing the manuscript, prepare a one-page scientific core. If you cannot answer the following questions clearly, the work is not yet ready to be written as a journal paper.

- What real research problem are you addressing?
- Why is the problem important beyond your particular application?
- What have recent researchers already attempted?
- What exact limitation remains unresolved?
- What is your new scientific idea?
- How is your method different from the strongest relevant existing approaches?
- What claims will you make?
- What experiment can test each claim?
- What baseline will you compare against?
- What quantitative evidence will support your conclusion?
- What can another AR/VR researcher learn from your findings?

One-sentence novelty test

Complete this sentence precisely: 'Unlike existing approaches that _____, the proposed work _____, enabling _____. ' If this sentence is vague, the novelty is probably vague.

3. Title and Keywords

The title should communicate the scientific contribution, not merely the application domain. It should normally combine the proposed idea, the AR/VR context, and the research objective.

Avoid / Weak	Use / Strong scientific style
AR-Based Education System	Adaptive Augmented Reality Guidance for Reducing Cognitive Load and Improving Procedural Task Accuracy

Avoid / Weak	Use / Strong scientific style
VR Medical Training Application	Performance-Aware Virtual Reality Training for Adaptive Procedural Skill Acquisition

Use 4-7 focused keywords. Include the main technology (AR/VR/XR), the method or interaction mechanism, the application/research context, and important evaluation concepts. Do not repeat only words already obvious from the title.

4. Abstract

The abstract should be a compact scientific argument. Write it after completing the results, even though it appears first in the paper.

1. Problem context: What important limitation or challenge motivates the study?
2. Research gap: What is missing in existing AR/VR research?
3. Proposed contribution: What new framework, model, algorithm, interaction technique, or experimental design is introduced?
4. Evaluation: How was it tested? Mention participants/dataset/conditions/baselines when relevant.
5. Key quantitative results: Report the most important numerical findings. Avoid vague statements such as "better performance".
6. Scientific implication: State what the evidence demonstrates and why it matters.

Abstract quality check

A reviewer should understand the problem, gap, novelty, evaluation, main result, and contribution without reading the full paper.

5. Introduction

The Introduction should move from the important problem to the exact gap and finally to your contribution. Avoid one or two pages of generic statements about AR/VR being popular or emerging.

Paragraph 1 - Problem and significance: Explain the real technical, human, clinical, educational, industrial, safety, or interaction problem. Use evidence and recent literature where appropriate.

Paragraph 2 - Why AR/VR is relevant: Explain what immersive or spatial computing enables for this problem. Do not assume AR/VR is automatically superior.

Paragraph 3 - Current state of research: Summarize the strongest relevant research directions and what they have already achieved.

Paragraph 4 - Research gap: Identify precise unresolved limitations. This should be specific, evidence-based, and directly linked to your proposed method.

Paragraph 5 - Proposed solution: Introduce the proposed framework/method and explain the central idea in simple scientific language.

Paragraph 6 - Novelty and contributions: State what is genuinely new and list 3-4 contributions that are later verified in the paper.

Contribution rule

Every contribution listed in the Introduction must be supported later by a method, experiment, analysis, or result. Do not claim novelty that is never evaluated.

6. Related Work and Research Gap

The Related Work section should synthesize research, not list papers one after another. Organize the literature by themes that directly lead to the research gap.

Possible AR/VR themes include: immersive training systems; adaptive XR; multimodal interaction; hand/gaze/voice interaction; spatial guidance; embodiment and avatars; presence and immersion; cybersickness; collaborative XR; learning and retention; haptics; XR accessibility; registration/tracking; AI-enabled XR; digital twins; or domain-specific XR applications.

Avoid / Weak	Use / Strong scientific style
Author A developed a VR system. Author B proposed another VR system. Author C used AR for training.	Prior immersive training systems demonstrate improved access to repeatable practice; however, many rely on fixed guidance and evaluate only short-term task performance. Adaptive XR studies address personalization, but often lack controlled comparisons linking adaptation to objective task outcomes. This leaves uncertainty about whether real-time adaptive guidance provides measurable benefit over conventional VR training.

End the section with a research-gap synthesis. The final paragraph should make the proposed study feel necessary rather than optional.

7. Research Questions and Hypotheses

Research questions convert a software project into an investigation. Each question must be answerable through data.

- RQ1: Does the proposed method improve task-completion efficiency compared with a conventional AR/VR baseline?
- RQ2: Does the proposed interaction or adaptation method reduce error rate or improve accuracy?
- RQ3: How does the method affect cognitive workload, usability, presence, or cybersickness?
- RQ4: Which component of the proposed system contributes most to the observed improvement?

Use hypotheses when suitable. Example: H1 - Participants using the adaptive VR condition will complete the task faster than participants using fixed-guidance VR. H2 - The adaptive condition will reduce error rate without significantly increasing workload.

Alignment rule

Every RQ/Hypothesis -> one or more measurements -> one statistical/analytical result -> one interpretation in the Discussion.

8. Proposed Method / Framework / System

This section should explain the scientific method behind the system. Do not turn it into a software tutorial. Include only implementation details that are required to understand, reproduce, or evaluate the contribution.

8.1 System overview

Provide a research-quality architecture figure. A typical flow may include user/sensor input -> state/feature extraction -> proposed decision or inference module -> AR/VR rendering/feedback -> updated user behaviour.

8.2 Inputs and sensing

Define the data used: head pose, gaze, hand tracking, controller motion, gestures, voice, task events, physiological signals, environmental state, or other inputs.

8.3 Core proposed mechanism

Explain the novel algorithm, interaction technique, adaptation logic, model, optimization method, guidance strategy, or design principle. Use equations/pseudocode where appropriate.

8.4 Output and feedback

Explain how the method changes the immersive environment, task difficulty, guidance, visualization, interaction, or feedback.

8.5 Design rationale

Explain why the chosen mechanism is expected to solve the identified research gap. Link design choices to theory or prior evidence.

8.6 Computational and runtime considerations

If relevant, report latency, update rate, inference cost, frame rate, tracking frequency, or hardware requirements.

Implementation details to avoid unless scientifically relevant

Screenshots of Unity Inspector panels, step-by-step scene creation, asset import instructions, code screenshots, menu navigation, and routine UI configuration.

9. Research Methodology and Experimental Design

The methodology must allow another researcher to understand exactly how the study was conducted. It must be designed to test the paper's claims, not merely demonstrate the application.

Participants: Report sample size, recruitment, age range, relevant demographics, prior AR/VR experience, inclusion/exclusion criteria, expertise level, and compensation if applicable.

Study design: Specify within-subject, between-subject, mixed design, repeated measures, crossover, or other design. Explain randomization/counterbalancing.

Experimental conditions: Define the proposed condition and all comparison conditions clearly.

Tasks: Describe what participants performed, task complexity, number of trials, practice session, time limits, and success criteria.

Hardware: Report headset/display, controllers, trackers, sensors, computer/GPU/CPU where performance matters, and relevant calibration.

Software: Report engine/framework versions, SDKs, libraries, models, and major system settings required for replication.

Procedure: Explain consent, briefing, training, calibration, task sequence, rest periods, questionnaires, interviews, and debriefing.

Data collection: State exactly what logs, measurements, questionnaires, sensor streams, or task outcomes were recorded.

Ethics: For human-subject research, follow institutional requirements for ethics approval/clearance, consent, privacy, and data handling.

10. Baselines and Comparative Evaluation

A paper cannot claim improvement without a meaningful reference point. Select baselines that allow the reviewer to judge whether your contribution matters.

- A conventional/non-adaptive AR or VR version of your own system.
- A standard controller-based or menu-based interaction method.
- A strong published method that can be fairly reproduced.
- A traditional desktop/physical/manual method when the research question requires it.
- A simpler component-level baseline to isolate the effect of the new technique.

Fair comparison requirements

Keep tasks, hardware, datasets, exposure time, participant groups, and measurement conditions as comparable as possible. Explain unavoidable differences. Do not compare your optimized method against an intentionally weak baseline.

11. Metrics and Measurement Instruments

Choose metrics because they answer research questions. Do not add many metrics simply to make the paper appear extensive.

Evaluation dimension	Example measures
Task efficiency	Completion time, throughput, interaction count, path length
Task quality	Accuracy, error rate, success rate, precision, recall, recognition accuracy
Learning/training	Pre/post-test gain, retention, transfer, procedural errors
System performance	FPS, motion-to-photon latency, tracking error, registration error, response time
User experience	Usability, satisfaction, engagement, ease of interaction
Immersion	Presence, embodiment, agency, spatial awareness
Human factors	Cognitive workload, fatigue, cybersickness, discomfort
Behavioural/physiological	Gaze behaviour, motion, heart rate or other justified signals

For subjective constructs, prefer validated questionnaires/scales where appropriate. If you create a custom questionnaire, justify it and report how items were developed and analysed.

12. Statistical Analysis

Do not reduce a user study to percentages such as "80% liked the system." Select analysis according to your design, data type, assumptions, and hypotheses.

- Report descriptive statistics such as mean, median, standard deviation, interquartile range, or confidence intervals as appropriate.
- Use suitable inferential tests (for example, paired/independent t-tests, ANOVA/repeated-measures ANOVA, regression, mixed-effects models, or appropriate non-parametric alternatives).
- Check assumptions when required and state how violations were handled.
- Report effect sizes where appropriate, not only p-values.
- For multiple comparisons, use an appropriate correction when necessary.
- For qualitative data, explain the coding/thematic analysis procedure and how reliability or consensus was established.

Avoid / Weak	Use / Strong scientific style
The proposed system is better because $p < 0.05$.	The adaptive condition reduced completion time relative to fixed-guidance VR; the statistical test indicates that the difference is unlikely to be explained by random variation, while the effect size indicates the practical magnitude of the improvement.

13. Results

The Results section should report evidence without overstating interpretation. Structure it according to the research questions or evaluation dimensions.

- RQ1 - task performance results
- RQ2 - accuracy/error results
- RQ3 - workload/usability/presence/cybersickness results
- RQ4 - component/ablation results
- System-level performance results, if relevant

Use figures and tables to show comparisons clearly. Report exact values needed to support claims. State non-significant findings as clearly as significant ones. Do not hide negative or unexpected results.

Important

Application screenshots demonstrate appearance, not scientific effectiveness. They should not dominate the Results section.

14. Ablation, Sensitivity, Robustness and Generalization

If the proposed system contains several new components, show which component actually contributes to performance. This is especially important for AI-enabled, adaptive, multimodal, or algorithmic XR work.

- Ablation: remove one component at a time (e.g., no adaptation, no gaze, no contextual guidance).
- Sensitivity: vary important thresholds/parameters and report stability.
- Robustness: test under different lighting, tracking quality, user movement, task complexity, device conditions, or data noise where relevant.
- Generalization: test different users, tasks, scenes, devices, datasets, or expertise levels where feasible.
- Failure cases: show when the method performs poorly and explain why.

Strong scientific question

Do not ask only: 'Does the complete system work?' Also ask: 'Why does it work, when does it fail, and which part creates the improvement?'

15. Discussion

Discussion is where you explain the scientific meaning of the results. Do not simply repeat values already shown in tables.

- Explain why the proposed method produced the observed effect.
- Relate the findings back to each research question and hypothesis.
- Compare the findings with previous studies and explain agreements or differences.
- Discuss practical significance, not only statistical significance.
- Identify unexpected findings and plausible explanations.
- Explain implications for future AR/VR system design or research theory.
- Avoid claiming generality beyond the tested population, hardware, task, or dataset.

Avoid / Weak	Use / Strong scientific style
The proposed method achieved 18% lower completion time.	The reduction in completion time suggests that context-sensitive visual guidance decreased unnecessary visual search. The effect was strongest among novice users, indicating that adaptive guidance may be most valuable during early skill acquisition.

16. Limitations, Threats to Validity and Ethics

A strong paper openly states limitations. A limitation section increases credibility when it is specific and connected to future validation.

- Small or homogeneous participant sample.
- Testing only university students instead of target professionals.
- Short-term experiment without long-term retention/transfer testing.
- Single headset, sensor, device, or laboratory environment.

- Possible learning/order effects.
- Simulator sickness or fatigue influencing performance.
- Novelty effect: users may rate a new XR system highly because it is new.
- Limited task complexity or dataset diversity.
- Potential model/data bias in AI-enabled XR.
- Privacy risks when collecting gaze, motion, audio, video, biometrics, or spatial maps.

For human studies, students must follow the applicable institutional ethics process before data collection where required. Consent, privacy, anonymization, and secure storage must be handled properly.

17. Reproducibility and Implementation Details

Provide enough technical detail to reproduce the research, while removing routine development history.

- Software engine and version; XR SDK/runtime and version.
- Hardware/device model and relevant settings.
- Tracking/calibration procedure.
- Model architecture, training settings, hyperparameters, and data split for AI/ML components.
- Task parameters, adaptation rules, thresholds, timing, and feedback logic.
- Source code/repository or supplementary material when permitted.
- Data availability and any restrictions.
- Random seeds or repeated-run protocol for computational experiments where relevant.

18. Figures and Tables

Figures should carry scientific information. Use a consistent visual style and ensure that every figure is readable when printed or viewed at journal-column width.

- Figure 1: Research motivation or conceptual overview (only if useful).
- Figure 2: Proposed system/framework architecture.
- Figure 3: Key interaction/adaptation/algorithm flow.
- Figure 4: Experimental setup or task conditions.
- Figures 5+: Results with appropriate error bars/uncertainty.
- Table 1: Critical comparison of related work and identified gap.
- Table 2: Experimental conditions/participants/metrics where useful.
- Table 3+: Quantitative comparison, ablation, or robustness results.

Avoid

Decorative screenshots, unreadable architecture diagrams, screenshots containing tiny UI text, 3D renders that do not support a scientific claim, and figures copied from software interfaces.

19. Conclusion

The Conclusion should answer the research problem using the evidence obtained. It should not introduce new experiments or repeat the entire paper.

1. Restate the research problem briefly.
2. State the proposed method/contribution.
3. Report the most important quantitative finding(s).

4. State the scientific insight or implication.

5. Mention the most important future research direction arising from the limitations.

Avoid / Weak	Use / Strong scientific style
We successfully developed the VR application and it can be useful in the future.	This study introduced a performance-aware VR guidance framework and demonstrated that adaptive guidance can reduce task errors and completion time relative to fixed-guidance VR while maintaining acceptable workload. The findings support adaptive guidance as a promising design direction for procedural immersive training, subject to validation with larger and more diverse professional populations.

20. References and Literature Quality

A Q1-target manuscript must be grounded in strong, recent, directly relevant literature. Do not collect references only to increase the count.

- Prioritize high-quality peer-reviewed journal/conference research relevant to the exact method and evaluation problem.
- Include recent literature to demonstrate awareness of the current state of the field, together with important foundational studies.
- Use primary sources rather than citing secondary summaries when the original research is available.
- Do not cite unrelated papers simply because they contain the words AR or VR.
- Use references critically: identify what each line of research does well and what limitation motivates your study.
- Verify every citation, author list, title, year, DOI, and claim before submission. Do not include fabricated references.

21. Scientific Writing Style

Write as researchers, not as developers describing a college project. Use precise, evidence-based language.

Avoid / Weak	Use / Strong scientific style
Our project has many unique features.	The proposed framework introduces two specific mechanisms: performance-aware adaptation and gaze-informed contextual guidance.

Avoid / Weak	Use / Strong scientific style
The results are very good.	The proposed condition reduced median task error by X% relative to the baseline.

Avoid / Weak	Use / Strong scientific style
Users felt the system was excellent.	Participants reported higher usability scores in the proposed condition; the difference was analysed using the specified statistical test.

Avoid / Weak	Use / Strong scientific style
No one has done this before.	To the best of our review, prior studies have not jointly evaluated X and Y under Z conditions; this gap is supported by the literature synthesis.

Avoid / Weak	Use / Strong scientific style
This proves the method is best.	Within the tested conditions, the proposed method outperformed the selected baselines on the specified metrics.

Tone

Avoid exaggerated words such as revolutionary, perfect, best, groundbreaking, highly efficient, and state-of-the-art unless they are justified by direct evidence and appropriate comparison.

22. Recommended Paper Layout

The exact section names should follow the target journal, but the following structure is suitable for planning a strong AR/VR research manuscript.

Section	Purpose
Title / Abstract / Keywords	Concise research identity
1. Introduction	Problem, gap, novelty, contributions
2. Related Work	Critical synthesis and research gap
3. Research Questions / Hypotheses	What the study will test
4. Proposed Method / Framework	Scientific mechanism and architecture
5. Methodology / Experimental Design	Participants/data, tasks, conditions, procedure, metrics
6. Results	Evidence organized by RQs
7. Ablation / Robustness / Additional Analysis	Why the method works and when

Section	Purpose
8. Discussion	Interpretation, implications, comparison with prior work
9. Limitations / Threats to Validity	Scope and constraints
10. Conclusion	Main finding and future direction
Declarations / Data / Code / Ethics	As required by the target journal
References	Verified, relevant literature

23. Mandatory Submission Checklist

Before sending the manuscript for faculty review, every student/team should confirm all items below.

- ☐ The title reflects a research contribution, not only an application.
- ☐ The paper identifies a specific research problem and evidence-based research gap.
- ☐ The novelty can be explained clearly in one or two sentences.
- ☐ The Introduction contains 3-4 defensible contributions.
- ☐ Recent and high-quality related work is critically compared.
- ☐ Research questions/hypotheses are explicitly stated or clearly implied by the study design.
- ☐ The proposed method contains a scientific mechanism, not just software features.
- ☐ At least one meaningful baseline/comparison is included.
- ☐ The experiment directly tests the main claims.
- ☐ Participant/data details and procedures are sufficiently reported.
- ☐ Metrics are justified and connected to research questions.
- ☐ Appropriate statistical or analytical methods are used.
- ☐ Results contain quantitative evidence and uncertainty where appropriate.
- ☐ Ablation/robustness/generalization is included when required by the contribution.
- ☐ The Discussion explains why the results occurred and what they mean.
- ☐ Limitations and threats to validity are stated honestly.
- ☐ Ethics/privacy requirements are satisfied for human or sensitive data.
- ☐ Figures and tables support scientific claims and are publication quality.
- ☐ The conclusion reports what was learned, not merely what was built.
- ☐ All references are genuine, verified, relevant, and correctly formatted.
- ☐ The manuscript has been proofread for grammar, consistency, terminology, and unsupported claims.

Final Standard Expected from Every Student Paper

The manuscript should demonstrate all of the following

- A clear and important research problem.
- A specific gap in recent literature.
- A genuine and defensible novelty.
- A technically clear proposed method.
- A rigorous experiment designed around research questions.
- Fair comparison with meaningful baselines.
- Appropriate quantitative/statistical evidence.
- Scientific interpretation, not just result reporting.
- Honest limitations and reproducibility details.
- A contribution that another AR/VR researcher can learn from.

A strong paper is not a longer project report. It is a focused scientific argument supported by reproducible evidence.